Six Iterated Integrals over a Tetrahedron

Stewart, *Calculus: Early Transcendentals*, §15.7, Exercise 32

Problem

Express the integral

$$\iiint_E f(x, y, z)\, dV$$

as an iterated integral in six different ways, where E is the solid bounded by the surfaces

$$x = 2, \qquad y = 2, \qquad z = 0, \qquad x + y - 2z = 2.$$

Part (a)

What the symbols mean

- E is the solid region of space. The integral adds the values of f over all points of E.
- f is a given function of three variables. In every answer below, f means $f(x, y, z)$.
- dV is the volume element, the product of the three differentials. Its order must match the order of the integral signs, so $dV = dz\, dy\, dx$ belongs to the order $dz\, dy\, dx$.
- An *iterated integral* is a chain of three integrals. The outer one runs over a fixed interval, the middle one over an interval that may depend on the outer variable, and the inner one between two surfaces written as functions of the two variables that are already fixed.
- The *shadow* of E on a coordinate plane is the set of points (u, v) such that the line through (u, v) parallel to the remaining axis meets E. The shadow gives the limits of the two outer integrals.
- *Fubini's theorem* states that for a continuous f the value of the integral does not depend on the order of integration, provided that every order used describes the same solid with correctly oriented limits. That theorem is why six different orders can all be correct.
- *Orientation rule.* A chain of limits of the form $a(u, v) \le w \le b(u, v)$ describes a solid only when $a(u, v) \le b(u, v)$ at every point of the shadow. If $a > b$ somewhere on the shadow, the inner integral runs backward on that part and *subtracts* instead of adds. Section "The base triangle and the trap" shows what that rule prevents.

Description of the solid

The four surfaces are planes, so E is a tetrahedron. Every vertex of a tetrahedron is the meeting point of three of its four faces, so each choice of three planes gives one vertex: $x = 2$, $y = 2$, $z = 0$ give $C = (2, 2, 0)$; $x = 2$, $y = 2$, $x + y - 2z = 2$ give $2 + 2 - 2z = 2$, so $z = 1$ and $D = (2, 2, 1)$; $x = 2$, $z = 0$, $x + y - 2z = 2$ give $2 + y = 2$, so $y = 0$ and $A = (2, 0, 0)$; $y = 2$, $z = 0$, $x + y - 2z = 2$ give $x + 2 = 2$, so $x = 0$ and $B = (0, 2, 0)$.

A point of the tetrahedron must lie on the correct side of each of the four planes. Test the interior point $\left(\frac{3}{2}, \frac{3}{2}, \frac{1}{4}\right)$, which is the centroid of the four vertices. It satisfies $x \le 2$, $y \le 2$, $z \ge 0$ and $x + y - 2z = \frac{3}{2} + \frac{3}{2} - \frac{1}{2} = \frac{5}{2} \ge 2$. Therefore

$$E = \{(x, y, z) : x \le 2,\ y \le 2,\ z \ge 0,\ x + y - 2z \ge 2\}.$$

The last inequality is equivalent to

$$z \le \frac{x + y - 2}{2},$$

so the plane $x + y - 2z = 2$ is the roof over the floor $z = 0$. Two further inequalities hold automatically. From $x \ge 2 + 2z - y$ and $y \le 2$ and $z \ge 0$ follows $x \ge 0$, and the same argument gives $y \ge 0$. So the solid lies in the first octant, and the four faces are

$$z = 0 \text{ (triangle } ABC), \qquad x = 2 \text{ (triangle } ACD),$$
$$y = 2 \text{ (triangle } BCD), \qquad x + y - 2z = 2 \text{ (triangle } ABD).$$

Figure 1 shows the solid and its four faces.

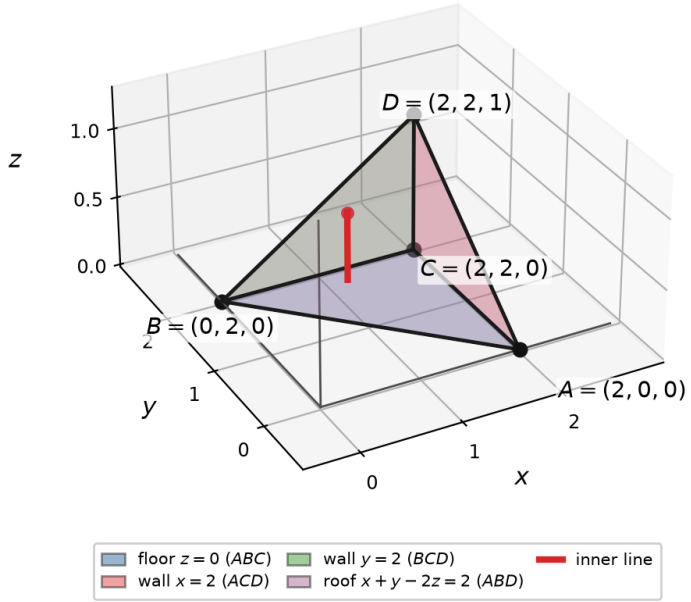


Figure 1: The solid E is the tetrahedron $ABCD$ with $A = (2, 0, 0)$, $B = (0, 2, 0)$, $C = (2, 2, 0)$ and $D = (2, 2, 1)$. The blue triangle ABC is the floor $z = 0$. The red triangle ACD is the wall $x = 2$. The green triangle BCD is the wall $y = 2$. The violet triangle ABD is the slanted roof $x + y - 2z = 2$. The red segment is one line of the inner integration: it starts on the floor at the point $(1.2, 1.8, 0)$ and stops on the roof at $z = (1.2 + 1.8 - 2)/2 = 0.5$.

The base triangle and the trap

Set $z = 0$ in the four inequalities. The floor of the solid is

$$T = \{(x, y) : 0 \le x \le 2,\ 0 \le y \le 2,\ x + y \ge 2\}.$$

This is the triangle with vertices $A = (2, 0)$, $B = (0, 2)$ and $C = (2, 2)$. Its right angle is at C, and both legs have length 2, so its area is 2. The point (x, y) of T carries the vertical segment

$$0 \le z \le \frac{x + y - 2}{2},$$

which has length 0 on the edge $x + y = 2$ and length 1 at $C = (2, 2)$.

Now consider the chain that ignores the edge $x + y = 2$, that is, the outer limits $0 \le x \le 2$, $0 \le y \le 2$ over the whole square, with the same inner limits. That chain is **wrong**. For the constant function $f = 1$ it gives

$$\int_0^2 \int_0^2 \frac{x + y - 2}{2}\, dy\, dx = \frac{1}{2}\left(\int_0^2 \int_0^2 x\, dy\, dx + \int_0^2 \int_0^2 y\, dy\, dx - 2 \cdot 4\right) = \frac{1}{2}(4 + 4 - 8) = 0.$$

The volume of E is $2/3$, as the next subsection proves twice. A limit chain that returns 0 for $f = 1$ cannot describe a solid of volume $2/3$, so the chain is not a description of E. This is the contradiction.

The cause is visible in Figure 2. The square $0 \le x \le 2$, $0 \le y \le 2$ is the union of the triangle T, where $x + y \ge 2$, and the triangle $x + y < 2$, which does not belong to the solid. On the second triangle the upper limit $(x + y - 2)/2$ is *negative*, so it lies below the floor $z = 0$. There the inner integral runs from 0 down to a negative number. An integral with the upper limit below the lower limit is the negative of the integral in the other direction, so the phantom triangle is counted with a minus sign and cancels half of the solid:

$$\int_T \frac{x + y - 2}{2}\, dA = \frac{2}{3}, \qquad \int_{\text{phantom}} \frac{x + y - 2}{2}\, dA = -\frac{2}{3}, \qquad \frac{2}{3} - \frac{2}{3} = 0.$$

The map $(x, y) \mapsto (2 - x, 2 - y)$ carries the phantom triangle onto T and changes the sign of $x + y - 2$, which is why the two numbers have equal size.

Two remarks make the point precise. First, as a *set of points* the wrong chain does not add points to E: the inequalities $0 \le z \le (x + y - 2)/2$ with $x + y < 2$ have no solution, because they demand a number that is both nonnegative and negative. The defect is not a larger point set but the *orientation* of the inner integral on part of the shadow. Second, the wrong chain fails for other functions too: for $f = z$ it returns $1/3$ instead of $1/6$, and for $f = xy$ it returns $4/3$ instead of $22/15$.

The cure is to take the shadow of E on the xy-plane, which is T and not the square. Every order below uses a shadow on which the inner lower limit is at most the inner upper limit. The script `part1_step3.py` tests this property directly: it samples 400,000 random points, finds 0 reversed inner intervals for all six orders, and finds 78,537 reversed inner intervals, that is 50.08% of the square, for the wrong chain.

Volume as a check

The tetrahedron has base area 2 and height 1, because the vertex $D = (2, 2, 1)$ stands 1 unit above the base plane $z = 0$. Hence

$$V = \frac{1}{3} \cdot 2 \cdot 1 = \frac{2}{3}.$$

A second route uses the sections drawn in the right panel of Figure 2. At height z the section is

$$T(z) = \{(x, y) : 2z \le x \le 2,\ 2z \le y \le 2,\ x + y \ge 2 + 2z\},$$

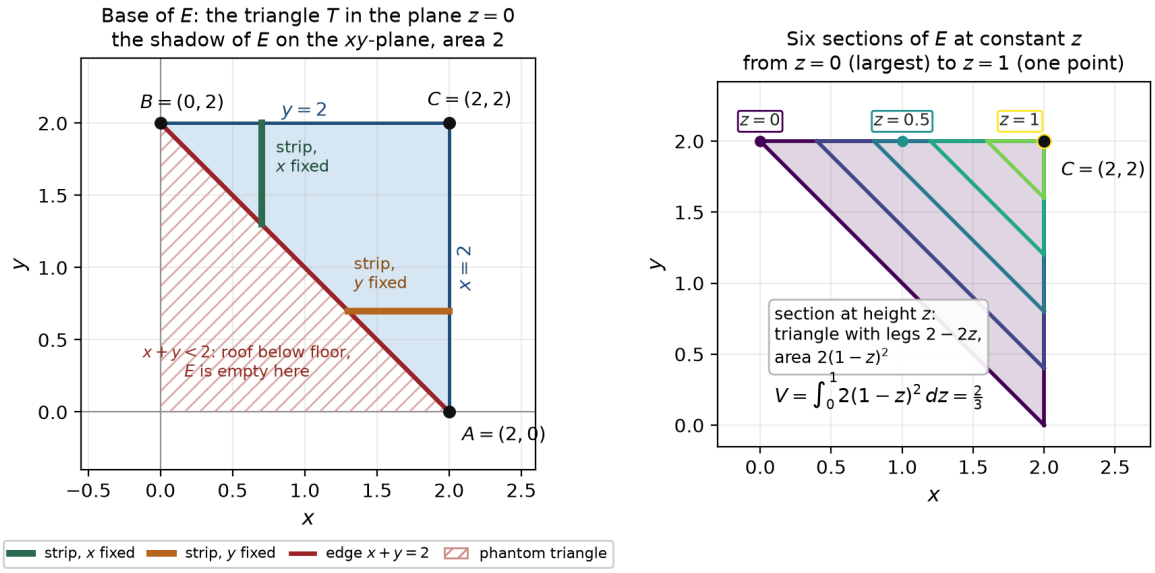


Figure 2: Left: the base T of E in the plane $z = 0$, the shadow of E on the xy-plane, area 2. The green segment is a strip with x fixed, from $y = 2 - x$ to $y = 2$. The orange segment is a strip with y fixed, from $x = 2 - y$ to $x = 2$. The hatched triangle $x + y < 2$ belongs to the square $0 \le x, y \le 2$ but not to E: there the roof $z = (x + y - 2)/2$ is below the floor $z = 0$, so the inner integral of the wrong chain runs backward. Right: the section of E at height z is a triangle with legs $2 - 2z$ and area $2(1 - z)^2$; the drawn levels are $z = 0, 0.2, \ldots, 1$, and at $z = 1$ the section is the point $C = (2, 2)$.

a right triangle with legs $2 - 2z$ and area $2(1 - z)^2$. Therefore

$$V = \int_0^1 2(1 - z)^2 \, dz = \left[-\frac{2(1 - z)^3}{3} \right]_0^1 = \frac{2}{3}.$$

A third route evaluates order (1) below with $f = 1$. The inner antiderivative is $(x - 2)y/2 + y^2/4$, so

$$\begin{aligned} \int_0^2 \int_{2-x}^2 \frac{x + y - 2}{2} \, dy \, dx &= \int_0^2 \left((x - 1) + \frac{(2 - x)^2}{4} \right) dx \\ &= 0 + \frac{1}{4} \cdot \frac{8}{3} = \frac{2}{3}, \end{aligned}$$

because $\int_0^2 (x - 1) \, dx = 0$ and $\int_0^2 (2 - x)^2 \, dx = 8/3$.

The six iterated integrals

The recipe for each order is the same. Pick the inner differential. The two outer variables then run over the shadow of E on their coordinate plane, and the inner limits come from the floor and the roof, solved for the inner variable. The base T can be read in two ways, with x fixed or with y fixed, and the shadow can be taken on any of the three coordinate planes. Together these choices give the six orders.

(1) Order $dz\,dy\,dx$. The outer variable is x, so x runs from 0 to 2. For a fixed x the shadow T gives $2 - x \le y \le 2$; this is the green strip of Figure 2. For a fixed (x, y) the vertical line enters E at the floor $z = 0$ and leaves it at the roof $z = (x + y - 2)/2$. The lower limit does not exceed the upper limit, because $x + y \ge 2$ on T.

$$\iiint_E f\,dV = \int_0^2 \int_{2-x}^2 \int_0^{(x+y-2)/2} f(x, y, z)\,dz\,dy\,dx.$$

(2) Order $dz\,dx\,dy$. This is the same solid with the two outer integrals exchanged. The outer variable is now y, and the shadow read with y fixed gives $2-y \le x \le 2$, the orange strip of Figure 2. Both outer limits are constants, so the exchange is legal, and the inner limits are unchanged.

$$\iiint_E f\,dV = \int_0^2 \int_{2-y}^2 \int_0^{(x+y-2)/2} f(x, y, z)\,dz\,dx\,dy.$$

(3) Order $dy\,dz\,dx$. Now y is the inner variable. The outer variable is x with $0 \le x \le 2$. The shadow on the xz-plane is

$$S_{xz} = \{(x, z) : 0 \le x \le 2,\ 0 \le z \le x/2\},$$

because $z \le (x + y - 2)/2$ and $y \le 2$ give $z \le x/2$. For a fixed (x, z) inside S_{xz} the line parallel to the y-axis enters E at the roof plane, where $y = 2 - x + 2z$, and leaves it at the wall $y = 2$. On S_{xz} the difference $2 - (2 - x + 2z) = x - 2z$ is nonnegative, so the limits are in the right order.

$$\iiint_E f\,dV = \int_0^2 \int_0^{x/2} \int_{2-x+2z}^2 f(x, y, z)\,dy\,dz\,dx.$$

(4) Order $dy\,dx\,dz$. The inner variable is again y, and the outer variable is now z. The solid occupies $0 \le z \le 1$. For a fixed z the shadow on the xy-plane is $T(z)$, the section of Figure 2, and the x-range of $T(z)$ is $2z \le x \le 2$. For a fixed (x, z) the line parallel to the y-axis enters at $y = 2 + 2z - x$ and leaves at $y = 2$. The difference $2 - (2 + 2z - x) = x - 2z$ is nonnegative on $T(z)$.

$$\iiint_E f\,dV = \int_0^1 \int_{2z}^2 \int_{2+2z-x}^2 f(x, y, z)\,dy\,dx\,dz.$$

(5) Order $dx\,dz\,dy$. The inner variable is x. The outer variable is y with $0 \le y \le 2$. For a fixed y the shadow on the xz-plane is

$$\{(x, z) : 0 \le x \le 2,\ 0 \le z \le y/2\},$$

because $z \le (x + y - 2)/2$ and $x \le 2$ give $z \le y/2$. For a fixed (x, z) the line parallel to the x-axis enters at the roof plane $x = 2 - y + 2z$ and leaves at the wall $x = 2$.

$$\iiint_E f\,dV = \int_0^2 \int_0^{y/2} \int_{2-y+2z}^2 f(x, y, z)\,dx\,dz\,dy.$$

(6) Order $dx\,dy\,dz$. The inner variable is x and the outer variable is z with $0 \le z \le 1$. For a fixed z the y-range of the section $T(z)$ is $2z \le y \le 2$, and the line parallel to the x-axis enters at $x = 2 + 2z - y$ and leaves at $x = 2$.

$$\iiint_E f\,dV = \int_0^1 \int_{2z}^2 \int_{2+2z-y}^2 f(x,y,z)\,dx\,dy\,dz.$$

Table 1 collects the three pairs of limits of each order. In every row the inner limits may use the two variables that are already fixed, the middle limits may use the outer variable only, and the outer limits are constants.

Table 1: The six orders of integration. Each row is a complete description of E, and each inner pair satisfies lower limit $\le$ upper limit on its whole shadow.

Order	Inner limits	Middle limits	Outer limits
$dz\,dy\,dx$	$0 \le z \le (x+y-2)/2$	$2-x \le y \le 2$	$0 \le x \le 2$
$dz\,dx\,dy$	$0 \le z \le (x+y-2)/2$	$2-y \le x \le 2$	$0 \le y \le 2$
$dy\,dz\,dx$	$2-x+2z \le y \le 2$	$0 \le z \le x/2$	$0 \le x \le 2$
$dy\,dx\,dz$	$2+2z-x \le y \le 2$	$2z \le x \le 2$	$0 \le z \le 1$
$dx\,dz\,dy$	$2-y+2z \le x \le 2$	$0 \le z \le y/2$	$0 \le y \le 2$
$dx\,dy\,dz$	$2+2z-y \le x \le 2$	$2z \le y \le 2$	$0 \le z \le 1$

Three integrals evaluated

Three functions are integrated over E: $f = 1$, $f = z$ and $f = xy$. The exact values are $2/3$, $1/6$ and $22/15$.

The volume, $f = 1$. The value is $2/3$. Section "Volume as a check" gives two geometric routes and evaluates order (1) in full.

The first moment, $f = z$. With order (1), the inner integral is $\int_0^{(x+y-2)/2} z\,dz = (x+y-2)^2/8$. Substitute $u = x + y - 2$, so $du = dy$; when y runs from $2 - x$ to 2, the new variable u runs from 0 to x. Then

$$\iiint_E z\,dV = \int_0^2 \int_0^x \frac{u^2}{8}\,du\,dx = \int_0^2 \frac{x^3}{24}\,dx = \frac{1}{24}\cdot\frac{16}{4} = \frac{1}{6}.$$

A second route uses a property of a tetrahedron: the average of a linear function equals its average over the four vertices. Here $\bar z = (0+0+0+1)/4 = 1/4$, so $\iiint_E z\,dV = V\bar z = \frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}$.

The mixed moment, $f = xy$. With order (1) the inner integral of xy over z gives $xy(x+y-2)/2$. With the same substitution $u = x + y - 2$ and $y = u + 2 - x$,

$$\iiint_E xy\,dV = \int_0^2 \int_0^x x\,(u+2-x)\,\frac{u}{2}\,du\,dx = \frac{1}{2}\int_0^2 \int_0^x \left(xu^2 + x(2-x)u\right)du\,dx.$$

The inner integral is $\frac{x^4}{3} + \frac{x^3(2-x)}{2}$. Hence

$$\iiint_E xy\,dV = \frac{1}{2}\int_0^2 \left(\frac{x^4}{3} + x^3 - \frac{x^4}{2}\right)dx = \frac{1}{2}\left(\frac{32}{15} + 4 - \frac{16}{5}\right) = \frac{1}{2}\left(\frac{32}{15} + \frac{4}{5}\right) = \frac{22}{15}.$$

A second route uses the closed formula for a tetrahedron with vertices $P_0, \dots, P_3$, $\int_E x_i x_j \, dV = \frac{V}{20}\big[(\sum_k P_k^i)(\sum_k P_k^j) + \sum_k P_k^i P_k^j\big]$. Here $\sum x = 6$, $\sum y = 6$ and $\sum xy = 0 + 0 + 4 + 4 = 8$, so $\int_E xy \, dV = \frac{2/3}{20}[36 + 8] = \frac{22}{15}$.

Verification

Four files check the answer. Each one rebuilds the limits from the definition of E; none of them reads the text above.

- `part1_step1.py` integrates all six orders exactly with `sympy`, for $f = 1$, $f = z$ and $f = xy$, and compares the results with reference values from the edge determinant, from the closed formulas for a simplex, and from a barycentric map of the unit 3-simplex. It also evaluates the wrong full square and its two halves.
- `part1_step2.py` computes the same six integrals with a nested Gauss–Legendre product rule (24 nodes per layer; all limits are affine, so the rule is exact for these polynomial integrands). It prints a convergence table for $n = 2, 3, 4, 6, 8, 24$ and runs a Monte Carlo check with 6,000,000 samples in the box $[0, 2] \times [0, 2] \times [0, 1]$.
- `part1_step3.py` tests the limits as sets of points. It draws 400,000 random points in $[-0.6, 2.6] \times [-0.6, 2.6] \times [-0.3, 1.3]$, keeps the points farther than 10^{-9} from every face, and compares the geometric test of E with the inequality chain of each order. It also counts, for every order, the points of the outer domain where the inner lower limit exceeds the inner upper limit.
- `part1_figure1.py` and `part1_figure2.py` draw Figures 1 and 2.

The exact values and the numerical values agree. Table 2 lists the six orders.

Order of integration	$f = 1$	$f = z$	$f = xy$
$dz\,dy\,dx$	0.666666666667	0.166666666667	1.466666666667
$dz\,dx\,dy$	0.666666666667	0.166666666667	1.466666666667
$dy\,dz\,dx$	0.666666666667	0.166666666667	1.466666666667
$dy\,dx\,dz$	0.666666666667	0.166666666667	1.466666666667
$dx\,dz\,dy$	0.666666666667	0.166666666667	1.466666666667
$dx\,dy\,dz$	0.666666666667	0.166666666667	1.466666666667
exact value	2/3	1/6	22/15

Table 2: Values of the six iterated integrals for the three test functions (Gauss–Legendre product rule, 24 nodes per layer). The spread over the six orders is 1.1×10^{-16} for $f = 1$, 3.6×10^{-16} for $f = z$ and 1.1×10^{-15} for $f = xy$, which is the level of the round-off error.

Monte Carlo with 6,000,000 samples gives $V = 0.666265 \pm 0.000608$, $\int_E z \, dV = 0.166714 \pm 0.000200$ and $\int_E xy \, dV = 1.465991 \pm 0.001430$, and each interval contains its exact value. The sampling box has volume 4, and the measured inside fraction 0.166566 matches $V/4 = 0.166667$.

Answer

$$\boxed{\begin{array}{ll} (1)\ \displaystyle\int_0^2\int_{2-x}^2\int_0^{\frac{x+y-2}{2}} f\,dz\,dy\,dx & (2)\ \displaystyle\int_0^2\int_{2-y}^2\int_0^{\frac{x+y-2}{2}} f\,dz\,dx\,dy \\ (3)\ \displaystyle\int_0^2\int_0^{\frac{x}{2}}\int_{2-x+2z}^2 f\,dy\,dz\,dx & (4)\ \displaystyle\int_0^1\int_{2z}^2\int_{2+2z-x}^2 f\,dy\,dx\,dz \\ (5)\ \displaystyle\int_0^2\int_0^{\frac{y}{2}}\int_{2-y+2z}^2 f\,dx\,dz\,dy & (6)\ \displaystyle\int_0^1\int_{2z}^2\int_{2+2z-y}^2 f\,dx\,dy\,dz \end{array}}$$

with $f = f(x, y, z)$ in every integrand. The solid has volume $V = \boxed{\frac{2}{3}}$ and the three exact values are

$$\iiint_E 1\,dV = \frac{2}{3}, \qquad \iiint_E z\,dV = \frac{1}{6}, \qquad \iiint_E xy\,dV = \frac{22}{15}.$$

How to solve problems of this kind

1. Name the surfaces and find the vertices first. Each vertex of a tetrahedron is the common point of three faces, so solve four systems of three linear equations.
2. Turn "bounded by" into inequalities, and test one interior point, for example the centroid of the vertices, to fix the direction of each inequality.
3. Find the shadow on the plane of the two outer variables. The shadow, not the bounding box, gives the outer limits. A box is correct only when the solid fills the box.
4. Check the orientation of every inner pair. The lower limit must be at most the upper limit on the whole shadow. If the roof dips below the floor somewhere, the shadow is too large: cut it. This is the trap of this exercise, and the wrong chain returns 0 for $f = 1$.
5. Give each variable its own range: outer limits are constants, middle limits may use the outer variable, and inner limits may use both outer variables.
6. Use the section test. For a fixed value of one variable, describe the section and compute its area. The area as a function of that variable gives the volume in one integral and confirms the orders that use it as the outer variable.
7. Check with $f = 1$ and with geometry. For a tetrahedron $V = \frac{1}{3}$(base area)(height), and all six orders must return that number.
8. Check one non-constant f by a second route. A linear function has its average at the average of the vertices, a quadratic monomial has the closed formula for a simplex, and quadrature or Monte Carlo gives a third route.
9. Test the limits with random points. Sample the bounding box, apply the geometric test of the solid and the inequality chain of each order, and require agreement on every point with no reversed inner interval.